BY ORDER OF THE COMMANDER
439TH AIRLIFT WING

439 AIRLIFT WING INSTRUCTION
21-108

1 SEPTEMBER 2026

Maintenance

CRASH, DAMAGED, DISABLED AIRCRAFT RECOVERY

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This instruction implements Department of the Air Force Policy Directive (DAFPD) 21-1, *Maintenance of Military Materiel*. This instruction extends the guidance of Department of the Air Force Instruction (DAFI) 21- 101_AFRC SUP 1, *Aircraft and Equipment Maintenance Management*, and 439th Airlift Wing Installation Emergency Management Plan (439 AW IEMP) 10-2. This instruction establishes individual responsibilities, restrictions, and documentation requirements for responsibilities and procedures necessary for Crashed, Damaged, Disabled Aircraft Recovery (CDDAR) in the Westover Air Reserve Base (WARB) area of responsibility (AOR) to include unit equipped and transient aircraft. Squadron commanders and maintenance supervisors are responsible for executing the group's CDDAR program in accordance with this instruction and Technical Order 00-80C-1, *Crashed, Damaged, Disabled Aircraft Recovery Manual*. All agencies involved with recovery operations will ensure compliance with this instruction to ensure a cooperative, coordinated response to CDDAR situations. The 439th Maintenance Group (439 MXG) Supervisors will monitor the CDDAR program and assist squadrons as necessary. This instruction requires collecting and maintaining information protected by the Privacy Act of 1974. System of records notice applies. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the DAF Form 847, *Recommendation for Change of Publication*; route DAF847 to 439 MXG/MXQ, 350 Hangar Ave., Box 21, Westover ARB, Chicopee, MA 01022. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with AFI 33-322, *Records Management and Information Governance*, and disposed of in accordance with the Air Force Records Disposition Schedule (RDS) located at **https://www.my.af.mil/gcss-af61a/afrims/afrims/**.

SUMMARY OF CHANGES

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1. Overview. The CDDAR Team must be prepared to respond to aircraft incidents for Westover ARB and surrounding areas. Primary considerations of the CDDAR team are to open the runway for operational use, to prevent secondary damage to disabled/incident aircraft and to preserve evidence for accident/mishap investigation.

2. Procedures/ Responsibilities.

2.1. The Incident Commander (IC) as determined by the 439 AW IEMP 10-2 is the person in charge at the incident until all emergency response actions are completed. Transfer of command to recovery organizations to include the CDDAR Team Chief takes place when hazard mitigation is complete. The Emergency Operations Center (EOC) under the EOC Director develops a recovery plan, which is approved by the Installation Commander before it is implemented. The EOC Director assures expedient, cooperative responses to CDDAR situations by 439 MXG, personnel, fire protection and other essential agencies.

2.2. All base organizations involved in crash recovery must be familiar with the 439 AW IEMP Plan 10-2 and the 439 AW Mishap Response Plan 91-204. The responsibilities of each essential base organization for CDDAR response are as follows:

2.2.1. 439th Base Civil Engineering, Fire Emergency Services (439 BCE/CEF) will assume IC duties at Fire, Hazardous Materials Emergency Planning and Response (HAZMAT) and Chemical, Biological, Radiological, Nuclear, hi-yield Explosives (CBRNE) on base accident/incident sites. The IC is the only point of contact at the accident/incident site. When the area is safe/secured, the IC responsibilities will be turned over to the CDDAR Team Chief.

2.2.2. 439th Security Forces Squadron (439 SFS) will report and coordinate with the IC to implement responsibilities to control the vicinity of the mishap site.

2.2.3. 439th Base Civil Engineering, Environmental Management (439 BCE/CEV) will advise the Emergency Operations Center (EOC) of hazardous material handling procedures. The CDDAR Team will assist the hazardous material response team members with clean-up, recovery and disposal procedures for all hazardous materials.

2.2.4. 439th Mission Support Group, Preventive Medicine (439 MSG/SGPB) will advise the EOC Director of the type(s) of personal protective equipment (PPE) required to perform recovery of an aircraft containing composite/hazardous materials. In consultation with the CDDAR team chief will perform annual survey of CDDAR responsibilities, duties, and recommend appropriate PPE to be available to eliminate possible health hazards every three years during the 439 MXS Aero Repair Health Risk Assessment. 439 MSG/SGPB will be available during crash recovery operation to advise on required occupational health medical evaluation/monitoring of personnel involved in CDDAR/clean-up operations; to include respirator, heavy metal testing or other necessary evaluations.

2.2.5. 439th Airlift Wing, Safety Office (439 AW/SE) will advise EOC Director of Safety hazards and concerns to ensure safe recovery operations. Provide CDDAR team chief with assistance and guidance in obtaining resources as needed to rectify unsafe conditions on site.

2.2.6. 439th Logistics Readiness Squadron, Vehicle Operations section (439 LGRV/PMI) will stand by to provide heavy equipment as directed by the EOC Director (**Attachment 2**). In the event of an off-base accident/incident, Vehicle Management will be requested to provide transport of CDDAR equipment along with Aircraft ground support equipment to the site.

2.2.7. The Contracting Office (439 CONF/PK) will procure specialized equipment not available at Westover at the request of the incident commander.

2.2.8. 439th Logistics Readiness Squadron, Fuels Management Flight (439LRS/LGRF) will stand by to provide de-fueling vehicle(s) as directed by the EOC Director.

2.2.9. 439th Airlift Wing, Command Post (439 AW/CP): Notification will first go through 439 AW/CP and they coordinate with 439th Civil Engineering Squadron, Explosive Ordnance (439 CES/CED) support and 439th Aircraft Maintenance Squadron (439 AMXS) concerning collection and disposal of all explosive hazards. Contact 439 AMXS Electronic Counter Measure section to assist with Flare/Chaff handling.

2.2.10. 439th Maintenance Group Commander (439 MXG/CC) will direct CDDAR operations for the accident/incident aircraft. Establish CDDAR capability and ensure resources and trained personnel are available to perform CDDAR responsibilities. Appoint 439th Maintenance Squadron (439 MXS) CDDAR Team Chief who will ensure vehicle/equipment requirements and recovery support equipment have 24-hour availability (**Attachment 2**).

2.2.11. 439th Maintenance Group, Maintenance Operations Center (MOC) will immediately establish and maintain communications to act as the aircraft maintenance focal point for receipt and transfer of all in-flight or ground emergency information within WARB and surrounding area. In addition, MOC shall run Quick Response Checklist 24 and contact all applicable agencies to respond as expeditiously as possible. When notified of a CDDAR, the senior controller will notify the 439 MXG/CC, key supervision, and the Crash Recovery Team Chief. Notification will include type of aircraft, location, amount of fuel and/or explosives on board and known extent of aircraft damage. Direct all personnel to stay clear of the recovery site unless assistance is requested by the incident commander.

2.2.12. 439th Maintenance Group, Quality Assurance Office (439 MXG/MXQ) will: Complete Aircraft Mishap Response Checklist 15 extracted from the 439 AW Mishap Response Plan 91-204. Provide the CDDAR Team Chief with guidance concerning aircraft weight and center of gravity computations for disabled/incident aircraft.

2.2.13. 439 AMXS Production Superintendents will: Initiate impoundment procedures, when directed. Direct ground movement of any home station or transient aircraft the EOC Director requests to be moved to facilitate CDDAR operations. Assist the CDDAR Team Chief as needed during recovery operations. Ensure a tow team is available to the CDDAR team for any assistance necessary to aid in the recovery operation. When directed, contact 439 LRS/LGRF through the EOC, to impound any fuel trucks used during ground refueling operations with a mishap aircraft at home station.

2.2.13.1. 439 MXG CDDAR Team Chief will identify team members and specific qualifications, ensure initial training is provided, and conduct recovery training exercises on an annual basis to maintain proficiency. Members of the crash recovery team shall be trained to respond to common incidents requiring CDDAR, i.e., aircraft departing prepared surfaces systems. Track initial and recurring training requirements in the Maintenance Information Systems.

2.2.13.2. Maintain up-to-date point of contact listing for all team members. The list will include contact numbers during duty and non-duty hours. The CDDAR team listing will also identify CDDAR team position, AFSC, and any special qualifications. A current copy of the listing will be available in MOC.

2.2.13.3. Conduct an annual briefing with all the agencies involved in the CDDAR process. This briefing will outline everyone's responsibilities for their respective areas. A tabletop exercise will also be conducted to check validity of telephone numbers, exercise checklists, as well as personnel capabilities. There will also be a discussion on possible responses to a variety of different scenarios. Maintain a CDDAR continuity book and review/update annually to assess personnel capabilities, exercise checklists, validity of home/business phone numbers, etc.

2.2.13.4. Ensure the CDDAR equipment is maintained and prepared to respond and deploy crash recovery equipment and personnel for major aircraft incidents.

2.2.13.5. Establish a crash recovery training program for team personnel and ensure sufficient team members are qualified in special handling procedures for any unique characterized hazards/materials for assigned aircraft. (e.g., depleted uranium, aircraft composite materials etc.) and document training. Personnel will accomplish annual training on assigned Mission Design Series (MDS) aircraft. Actual aircraft emergencies can be substituted for the required training.

2.2.13.6. Direct and coordinate CDDAR operations as instructed by the EOC through the EOC Director. The EOC will contact the CDDAR Team Chief when incident area is safe for recovery operations to begin. Prior to any recovery actions ensure any hazards, including toxic materials, munitions and radioactive materials are eliminated, and the aircraft and egress systems are made safe for recovery activities and investigation team members. Review and start the CDDAR Team Chief worksheets (Attachments **4, 5, 6, 7 and 8**). Evaluate the aircraft damage and prepare for recovery operations.

2.2.13.7. Ensure safety briefings are given prior to each new shift coming on duty detailing hazards to personnel and equipment. Maintain continuous communications with the 439 MXG/CC and MOC to keep them informed on the progress of the recovery operation, to include CDDAR team limitations. Document a locally devised CDDAR daily occurrence log to maintain a record of incident.

2.2.13.8. Ensure the crash recovery team members follow all applicable Technical Orders and safety procedures while conducting aircraft recovery.

3. Transient Aircraft Responsibilities.

3.1. CDDAR equipment on station is limited to assigned home station airframe. 439 MXG/CC will ensure additional special equipment and unique requirements for transient aircraft is obtained from applicable sources as directed by Aircraft Maintenance Division, Headquarters Air Force Reserve Command (HQ AFRC/A4M) (**Attachment 2**).

3.2. CDDAR team chief will contact owning organization to obtain necessary expertise and guidance from appropriate sources. All team members will be briefed on special requirements/health and safety concerns associated with transient aircraft recovery.

3.3. All base organizations will provide assistance with transient aircraft (including civilian aircraft) recovery operations as directed.

4. Safety Considerations.

4.1. All organizations must be aware of the following safety concerns present during crash recovery procedures.

4.1.1. All organizations must communicate and coordinate activities through the EOC during recovery operations.

4.1.2. Operational Risk Management practices will be utilized and are paramount in all decision-making.

4.1.3. Prior to any recovery actions ensure any hazards, including toxic materials, munitions, and radioactive materials are eliminated, and the aircraft and egress systems are made safe for recovery activities and investigation team members. All base organizations should be familiar with safety/health hazards associated with any unique characteristics/hazards/materials for assigned aircraft (e.g., Jet-A, depleted uranium, etc.) and document all required training.

CHRISTOPHER C. HOLLAND, Colonel, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFPD 21-1, *Maintenance of Military Materiel,* 21 February 2024

DAFI 21-101_AFRCSUP, *Aircraft and Equipment Maintenance Management,* 30 October 2024

DAFI 10-2501, *Emergency Management Program,* 16 October 2023

439 AW Installation Emergency *Management Plan 10-2,* 1 January 2025

439th Airlift Wing Mishap *Response Plan 91-204*, 2 January 2020

Adopted Forms

DAF Form 847, *Recommendation for Change of Product*

Abbreviations and Acronyms

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFSC—Air Force Specialty Code

AGE—Aerospace Ground Equipment

AMXS—Aircraft Maintenance Squadron

AWI—Airlift Wing Instruction

CBRNE—Chemical Biological, Radiological, Nuclear, hi-yield Explosives

CDDAR—Crashed, Damaged, Disabled Aircraft Recovery

DAFI—Department of the Air Force Instruction

DAFPD—Department of the Air Force Policy Directive

EOC—Emergency Operations Center

IC—Incident Commander

IEMP—Installation Emergency Management Plan

MDS—Mission Design Series

MOC—Maintenance Operations Center

MXG—Maintenance Group

MXS—Maintenance Squadron

OPR—Office of Primary Responsibility

PPE—Personal Protective Equipment

Office Symbols

439 AMXS—439th Aircraft Maintenance Squadron

439 AW/CC—Commander, 439th Airlift Wing

439 AW/CP—Command Post, 439th Airlift Wing

439 AW/SE—Safety Office, 439th Airlift Wing

439 BCE/CEF—Fire Emergency Services, 439th Base Civil Engineering

439 BCE/CEV—Environmental Flight, 439th Base Civil Engineering

439 CED—Explosive Ordnance Disposal, 439th Civil Engineering Squadron

439 CONF/PK—Contracting Office, 439th Airlift Wing

439 LGRF/PMI—Fuels Management Flight, 439th Logistics Readiness Squadron

439 LGRV/PMI—Vehicle Operations Section, 439th Logistics Readiness Squadron

439 MSG/SGPB—Preventive Medicine, 439th Mission Support Group

439 MXG/CC—Commander, 439th Maintenance Group

439 MXG/MXQ—Quality Assurance, 439th Maintenance Group

439 MXS—439th Maintenance Squadron

439 SFS—439th Security Forces Squadron

HQ AFRC/A4M—Aircraft Maintenance Division, Headquarters Air Force Reserve Command

Attachment 2

ADDITIONAL EQUIPMENT TO BE PURCHASED LOCALLY AS REQUIRED:

A2.1. Fixates, hold down solutions, polyacrylic acid, floor wax and other required items will be purchased as required.

A2.2. CDDAR Vehicles:

A2.2.1. Thirty-ton crane unit owned; 15-ton crane available through vehicle maintenance (24/7 availability).

A2.2.2. Flatbed truck, available through vehicle maintenance.

A2.2.3. All terrain forklift(s), available through vehicle maintenance.

A2.3. Ground Support Equipment:

A2.4. 439 MXS Aerospace Ground Equipment (AGE). Flight will supply, inspect, support and service all MC-7 air compressors, lighting units, heaters, air conditioning units, generators, jacking manifold and AGE Equipment as identified.

A2.5. CDDAR equipment, vehicle and support equipment list.

Table A2.1. CDDAR Equipment, Vehicle and Support Equipment List.

ITEM:	LOCATION:	ITEM:	LOCATION:
Air Bag Modules	Building 7040	Crash axe	CDDAR Trailer
Air Bag Inflation Manifolds	Building 7040 & CDDAR Trailer	Tyvek suits (Consumable)	CDDAR Trailer
Tow Bridles w/F2 trailers	Building 7040	Megaphone	CDDAR Trailer
Turfor Winches	CDDAR Trailer	40’ x 12’ lifting strap	CDDAR Trailer
Respirators, PPE	CDDAR Trailer	14’ spreader bar	Lift beam rack
60 Ton tripod jacks	Building 7040	Tow Vehicle, Tug	Maintained by 439 AMXS
30-ton tripod jacks	Building 7040	Response trailer tow vehicle	To be procured as a future purchase per vehicle ops. Vehicles located on base that can be utilized (fuel cell, 439 AMXS)
50-ton D-rings	CDDAR Trailer	Respirator Filters (Consumable)	CDDAR Trailer
30-ton D-rings	CDDAR Trailer	Floor Wax (Used as fixate, Consumable)	To be purchased when required
25-ton D-rings	CDDAR Trailer	28’ utility trailer (CDDAR Response Trailer)	Building 7040
17-ton D-rings	CDDAR Trailer		
18 ton 12’ straps	CDDAR Trailer		
15 ton 6’ straps	CDDAR Trailer		
Pickaxe	CDDAR Trailer		
Shovel	CDDAR Trailer		

Note: This list is an example of equipment and consumables but is not all encompassing. The equipment and consumables required will vary based on the nature of the incident. Items required for CDDAR response that are not possessed will be purchased, leased, rented, or borrowed as required.

Attachment 3

POINT OF CONTACT LIST – KEY PERSONNEL AND RESOURCES

Table A3.1. Point Of Contact List – Key Personnel and Resources.

1. AMC AIRCRAFT: 618th Air Operations Center (aka Tanker Airlift Control Center TACC), Scott AFB, Illinois, 1-800-247-6625
2. AFRC CRASH RECOVERY SUPERINTENDENTS:
a. HQ AFRC/A4MY - DSN: 497-1645 Comm: 478-327-1645
b. 4AF/A4MY - DSN: 447-7643 Comm: 951-655-7643
c. 10AF/LGMA - DSN: 739-5155 Comm: 817-782-5189
d. 22AF/A4M - DSN: 625-3853 Comm: 678-655-3853

Attachment 4

CRASH / DISABLED AIRCRAFT PRE-RESPONSE BRIEFING WORKSHEET

Table A4.1. Crash/Disabled Aircraft Pre-Response Briefing Worksheet.

1. Initial information on crash/mishap. What facts are known at present time?
2. Introduce all members of the team and what their specialty or expertise is.
3. Brief the sequence of events that will normally take place.
4. Name of the On Scene Commander.
5. Name of the DCG Maintenance Representative.
6. Name of the Crash Recovery Team Chief.
7. Await authorization from MOC or DCG Maintenance Rep to proceed to crash/mishap site.

Attachment 5

CRASH DISABLED AIRCRAFT RECOVERY WORKSHEET

Table A5.1. Crash Disabled Aircraft Recovery Worksheet.

(THIS WORKSHEET WILL BE ACCOMPLISHED BEFORE AIRCRAFT IS MOVED/LIFTED)

CRASH RECOVERY TEAM CHIEF WILL ENSURE:

1. Assemble Crash Recovery Team.
2. Brief the team on the current situation using the Pre-Response Briefing Checklist. (Attachment 5)
3. Check equipment and prepare for transportation to crash site.
4. Once authorization is given to access the crash site, use the Aircraft Data Checklist (page 3) to gather as much data as possible (e.g., aircraft weight, cargo, structural damage, etc.)
5. Use the Hazard Checklist (page 4) to identify what hazards exist.
6. Using data from the Aircraft Data Checklist and the Hazard Checklist, accomplish a risk assessment using Operational Risk Management (ORM).
   a. ID Risk. Identify the hazards.
   b. Assess the Risk. Risk is the probability and severity of loss linked to the hazard.
   c. Analyze Risk Control Measures. Investigate specific strategies and tools that reduce or eliminate risk.
   d. Make Control Decisions. Deal with risk by accepting, avoiding, reducing, spreading or transferring it.
   e. Risk Control Implementation. Develop implementation strategy.
   f. Supervise and review. Once controls are in place, the process must be scrutinized to determine its effectiveness.
7. Develop a recovery plan and assign duties, using the following factors:
   a. Aircraft condition (e.g., structural damage, gear up/down, weight, etc.)
   b. Terrain (e.g., level, sloped, muddy, snow, etc.)
   c. Weather (e.g., rain, wind, cold, etc.)
   d. Personnel and equipment available.
   e. Hazards identified using ORM
   f. Urgency of recovery (i.e., runway out of commission)
   g. Brainstorm all problems and encourage input from all personnel.
8. Before commencing with the recovery, give a safety briefing using the Safety Briefing Checklist (Attachment 8).
9. Fill out a Key Personnel Contact Information Sheet

Attachment 6

AIRCRAFT/SITE DATA WORKSHEET AIRCRAFT INFORMATION

Table A6.1. Aircraft/Site Data Worksheet Aircraft Information.

1. Aircraft Type and Model: ..... _____

2. Mishap Date: ..... _____

3. Time: ..... _____

4. Tail Number: ..... _____

5. Empty Weight: ..... _____

6. Total Fuel Load (current): ..... _____

7. Fuel Load by Tank: ..... _____

8. Cargo Load (weight): ..... _____

9. Cargo Type (pallets, rolling stock, vehicles, etc.): .. _____

10. Munitions on Board: ..... _____

11. Major Structure Condition: ..... _____

12. Landing Gear Intact: ..... _____

Additional aircraft information or comments: _____

_____

_____

_____

SITE INFORMATION

1. General Location:

_____

2. Exact Position: _____

3. Type Terrain: _____

4. Temperature: _____

5. Weather: _____

_____

6. Wind Speed and Direction: _____

7. Weather Forecast: _____

Additional site information or comments: _____

_____

_____

_____

Attachment 7

HAZARD IDENTIFICATION WORKSHEET

Table A7.1. Hazard Identification Worksheet.

YES/NO	COMMENTS
1. Explosives/Ammunition / Flares:	
2. Biohazard (blood / remains):	
3. Fuel:	
4. Toxic Chemicals:	
5. High Pressure Cylinders:	
6. Corrosive Agents:	
7. Ejection Seats:	
8. Ejection Hatches:	
9. Batteries:	
10. Damaged Tires:	
11. Composites:	
12. Review T.O. 00-105E-9:	

Attachment 8

CRASH/DISABLED AIRCRAFT RECOVERY SAFETY BRIEFING WORKSHEET

Table A8.1. Crash/Disabled Aircraft Recovery Safety Briefing Worksheet.

1. _____ All personnel involved with the recovery operation must have the following PPE.

   a. Steel toe boots
   b. Eye protection
   c. Ear protection
   d. Hard hat
   e. Safety vest
   f. Leather work gloves
   g. Reflective belt
   h. Whistle
   i. Dressed for weather conditions (e.g., cold, hot, rain, etc.)

2. _____ Except for an emergency, no team member will move any portion of the aircraft, unless it has been cleared by the On Scene Commander (MOSC) or the Team Chief. Each member will do only those tasks he/she was directed to do by the MOSC or the Team Chief. Team members will do assigned tasks until relieved by another team member.

3. ____ If for any reason, a team member cannot safely do his/her assigned tasks (too dangerous or too many people in the area) it is his/her responsibility to notify the team chief.

4. _____ Use spotters when backing the crash trailer, vehicles and equipment.

5. ____ Any team member can halt the lifting/recovery operation if an unsafe condition is noticed. Methods of alerting unsafe conditions will be by radios or whistles.

6. _____ Be familiar with the location of emergency equipment at the site, such as: fire extinguishers, eye wash units, first aid kits, safety data sheets (SDS).

7. _____ Establish a crash site evacuation plan in the event of a fire or other emergency.

8. _____ Establish procedures and lines of communication to various departments (fire dept, MOC, Safety, etc.) to assist in the event of an accident or injury.

9. _____ Be aware of your surroundings and the various hazards.

   a. Never be under an aircraft while it's being lifted unless manning a jack.
   b. Stay clear of guide/tow cables, chains or straps under tension.
   c. Take precautions to prevent falls from elevated areas (aircraft, stands, etc.)
   d. Always wear the appropriate PPE.
   e. Use proper lifting techniques.
   f. Take measures to avoid heat stroke, dehydration, sun burn, frost bite, etc. (drink plenty of water.